

Dataset Condensation

A PROJECT REPORT

Submitted by

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of

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in

COMPUTER SCIENCE AND ENGINEERING



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DECLARATION

I, **Likith S G (1RVU23CSE236)**, student third semester B.Tech in **Computer Science & Engineering**, at School of Computer Science and Engineering, **RV University**, hereby declare that the project work titled “Dataset Condensation” has been carried out by me and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science & Engineering** during the academic year **2024-2025**. Further, the matter presented in the project has not been submitted previously by anybody for the award of any degree or any diploma to any other University, to the best of our knowledge and faith.

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CERTIFICATE

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ABSTRACT

Machine learning models are typically trained on large-scale datasets containing high-quality images, a process that is both time-consuming and resource-intensive. This research explores Dataset Distillation, a technique designed to create smaller, synthetic versions of these datasets while maintaining comparable accuracy. Our study introduces innovative approaches that leverage Gradient Matching to align training gradients, ensuring that the synthetic datasets effectively replicate the performance of the original ones.

In addition to Gradient Matching, we review and integrate other techniques such as Trajectory Matching, Kernel Ridge Regression, and Parameter Pruning to present a comprehensive understanding of existing solutions in dataset distillation. We developed and evaluated four models—**MPS1**, **MPS1_with_GAN**, **MPS2**, and the enhanced **MPS2U**. **MPS2U**, an adaptation of **MPS2**, introduces optimizations such as early stopping and dynamic reinforcement learning to ensure improved training efficiency while maintaining high accuracy across diverse datasets like MNIST, Fashion-MNIST, CIFAR-10, and CIFAR-100. Its robust convolutional neural network (ConvNet) architecture is optimized for versatility, enabling effective learning on both simple and complex datasets.

A critical advancement in this work is the integration of **Mutual Information (MI)** for refined sample selection. MI-based scoring identifies the most informative samples, allowing the selection of high-quality subsets of 1,000 samples per dataset. By incorporating MI into **MPS2U**, the synthetic data generation process is further enhanced, ensuring the retention of essential data attributes while significantly reducing training time and computational overhead. This mutual information framework synergizes with reinforcement learning to dynamically improve sample quality and optimize dataset condensation.

Comparative results show that **MPS2U** and its MI-based extension outperform traditional methods like Gradient Matching in accuracy, training time, and resource efficiency, particularly for complex datasets. This research underscores the transformative potential of dataset condensation in modern machine learning workflows, offering a promising solution for scalable and efficient data management.

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1. INTRODUCTION

In the modern era of vast data generation and the increasing reliance on machine learning algorithms, the training process has become a significant bottleneck. Training models on large datasets often requires days or even weeks, consuming substantial computational resources and time. This challenge escalates as data volumes continue to grow. A promising solution to this problem is **Dataset Condensation**.

Dataset condensation involves creating smaller, synthetic versions of original datasets that retain comparable predictive performance. By using condensed datasets, training time and resource requirements can be significantly reduced, while maintaining similar levels of accuracy.

Several established techniques have been widely used in this domain, including:

- **Trajectory Matching:** This method aligns the training dynamics of models on synthetic data with those trained on real datasets, effectively replicating their training paths.
- **Kernel Ridge Regression (KRR):** By utilizing a kernel function, KRR maps features into a higher-dimensional space, making it easier to model complex, non-linear relationships in the data.
- **Gradient Matching:** This approach ensures that the gradients of the synthetic dataset match those of the original dataset during training, preserving similar learning behavior.
- **MPS2U (Memory Prediction Score to Update):** A novel strategy that evaluates and optimizes the alignment of synthetic data by quantifying its mutual predictive value with the original dataset. MPS2U iteratively refines the synthetic dataset to enhance its representational quality and predictive fidelity.

To further enhance dataset condensation, this research incorporates **Mutual Information** to identify and select the most informative samples from the original dataset. This ensures that the chosen samples capture the critical characteristics of each label. Once these representative samples are identified, **Reinforcement Learning** is employed to optimize and synthesize condensed datasets.

The reinforcement learning process, combined with the MPS2U strategy, focuses on maintaining the core attributes of the original data. As a result, machine learning models can be effectively trained on these distilled datasets, significantly reducing computational demands while preserving high performance.

2. Related work

Data distillation aims to reduce large datasets into smaller, more efficient subsets while preserving critical information, allowing for faster model training with minimal accuracy loss. Several key techniques have evolved over the years, addressing previous limitations and improving the efficiency of dataset distillation:

- **Trajectory Matching:**

Trajectory matching improves upon earlier distillation methods that focused on single training steps or entire parameter trajectories. This technique aligns segments of parameter trajectories from models trained on synthetic data with those trained on real data. By doing so, it ensures that synthetic data models achieve performance similar to those trained on real datasets. Recent experiments show that trajectory matching outperforms older methods, such as coreset selection, on datasets like CIFAR-10, achieving 71.5% accuracy with just 50 images per class. It also scales effectively to larger datasets like ImageNet, making it a significant advancement in the field.

- **Gradient Matching:**

Earlier techniques focused on directly matching model parameters, often resulting in high computational costs. Gradient matching addresses this by aligning gradients from synthetic and real datasets during training, ensuring that the parameters follow similar paths. This method improves optimization by making the training process faster and more memory-efficient. Unlike previous methods, gradient matching eliminates the need for costly parameter unrolling, enabling efficient training of large deep neural networks with minimal computational overhead.

- **Kernel Ridge Regression (KRR):**

KRR combines ridge regression with kernel functions to handle non-linear relationships in data, an improvement over traditional linear models. By mapping input data to higher-dimensional spaces using kernel functions such as linear, polynomial, or radial basis functions, KRR can capture complex patterns that linear regression cannot. This technique has been shown to perform well in regression tasks, particularly when dealing with non-linear data, offering a more flexible and efficient solution than earlier approaches.

- **Parameter Pruning:**

Parameter pruning improves upon traditional distillation methods by reducing the number of less important parameters in a model. This technique involves training multiple teacher

networks on the original dataset, saving their parameters, and then using those to initialize student networks. Non-essential parameters are pruned to help the student network focus on key patterns, reducing overfitting and improving training efficiency. This method enhances the distillation process by refining the dataset and model, making it particularly useful for applications like continual learning and neural architecture search.

These methods represent key advancements in dataset distillation, addressing limitations of earlier techniques by improving scalability, efficiency, and model performance. Each method contributes to reducing the computational costs of training machine learning models while maintaining or improving accuracy, providing a foundation for future research in the field.

3. METHODOLOGY

i. Research Design

The research utilizes an experimental framework focused on creating smaller, synthetic datasets from original large datasets. The aim is to determine if these smaller datasets can be used to train models efficiently without substantial loss of accuracy. Three distinct models—MPS1, MPS1_with_GAN, and MPS2—are developed and tested.

ii. Data Selection

In the research, four well-known datasets are used: **MNIST**, **Fashion-MNIST**, **CIFAR-10**, and **CIFAR-100** as test data. These datasets are well-established within machine learning and span a range of complexity to examine the success of dataset distillation methods. We sample **50 images per class** for the distillation process for each dataset. The original datasets are used as benchmarks for assessing the models' performance.

iii. Model Development

- **MPS1 Model:**
This model is a convolutional neural network (CNN) designed for classification tasks. It uses reinforcement learning to generate synthetic datasets. The reinforcement learning agent optimizes the selection of representative samples from the original dataset, reducing the dataset size without losing important features.
- **MPS1_with_GAN:**
This model builds upon MPS1 by including a **Generative Adversarial Network (GAN)**. The role of the GAN here is to generate a higher-quality version of those synthetic datasets by constructing data distributions more accurately, which effectively increases the generalization capability of the model with respect to the synthetic set.
- **MPS2 Model:**
MPS2 refines the earlier models by adding early stopping mechanisms to prevent overfitting and improve computational efficiency. MPS2 is designed to handle multiple datasets, including more complex ones like CIFAR-10 and CIFAR-100.
- **MPS2U Model:**
MPS2U was designed to enhance training efficiency, accuracy, and versatility across multiple datasets, focusing on reducing training time while maintaining high performance. MNIST and Fashion-MNIST benefitted from fewer epochs due to their

simpler structures. CIFAR-10 and CIFAR-100, with their increased complexity, required more epochs to adequately capture intricate patterns. A **Convolutional Neural Network (ConvNet)** was utilized to train on synthetic datasets from scratch. The architecture was optimized for effective learning and performance across the selected datasets.

- **Mutual Information:**

The **Mutual Information (MI)-Based method** builds upon MPS2U's principles, incorporating mutual information and reinforcement learning for enhanced sample selection. By employing mutual information, the method identified the top 1,000 samples with the highest scores, ensuring that only the most informative samples were used for training.

Reinforcement Learning Integration:

An RL agent was included to optimize sample selection based on model performance, creating a dynamic and iterative improvement process.

Adaptation of MPS2U:

The mutual information script was a direct adaptation of MPS2U, retaining its structural advantages while adding the capability of refined sample selection through mutual information.

iv. Training and Implementation

Each model is implemented using Python and relevant machine learning libraries such as **TensorFlow** and **PyTorch**. The models are trained on both the original and synthetic datasets. The training process involves optimizing the synthetic datasets using reinforcement learning (in MPS1) or GANs (in MPS1_with_GAN). MPS2 also incorporates early stopping mechanisms to minimize overfitting and reduce training time.

v. Performance Evaluation

The models are evaluated based on key metrics:

- **Accuracy:**

Compare the accuracy of the models trained with synthetic vs. original datasets. The exercise performed here is to ensure that models trained on synthetic datasets maintain accuracy levels similar to those trained on full datasets, despite the reduced dataset size.

- **Training Time:**

The time required to train each model is measured and compared. The expectation is that models trained on synthetic datasets will take significantly less time to train than those using full datasets.

- **Computational Efficiency:**

The use of computational resources, including **GPU usage** and **memory**, is monitored to assess the efficiency of the distillation techniques.

ix. Comparison with Existing Techniques

The results from the models developed in this research are compared with traditional dataset distillation techniques, particularly **Gradient Matching**. This comparison focuses on the trade-offs between computational efficiency, accuracy, and training time, providing insights into the advantages and limitations of each method.

4. IMPLEMENTATION

- **MPS2U Implementation:**

This was an adaptation of the previous script MPS2, which had employed reinforcement learning to create synthetic datasets.

- **ConvNet Training:**

A ConvNet was implemented to train on synthetic datasets, with its architecture designed to adapt easily to various dataset complexities.

- **Visualization of Synthetic Data:**

A directory function was added to save and visualize generated synthetic samples, facilitating analysis of model outputs.

- **Robust Input Handling:**

An improved input handling function was developed to preprocess and format data correctly for the ConvNet, enhancing the overall training pipeline.

- **Mutual Information-Based Implementation:**

- **Mutual Information Scoring:**

A scoring mechanism was introduced to rank samples based on mutual information, leading to the selection of the top 1,000 samples for training.

- **Reinforcement Learning:**

An RL agent optimized the sample selection process, enhancing training efficiency and effectiveness.

- **Integration with TensorFlow:**

The mutual information approach built upon MPS2U's framework, maintaining TensorFlow's advantages for efficient computation.

5. RESULT AND DISCUSSION

Metric	MPS2U	Mutual Info	MPS2	Gradient Matching
MNIST Test Accuracy	95%	88%	75%	98%
Fashion-MNIST Test Accuracy	77%	77%	58%	83%
CIFAR-10 Test Accuracy	31%	23%	31%	53%
CIFAR-100 Test Accuracy	12%	-	-	-
Average Training Time	5 minutes	10 minutes	20 minutes	4 hours
Computational Load	Low (Integrated GPU)	High	Low (Integrated GPU)	High

Figure 1: Performance metrics comparison of MPS2U and Mutual Info across four datasets: MNIST, FashionMNIST, CIFAR-10, and CIFAR-100. Benchmark algorithms MPS2 and Gradient Matching.



Figure 2: Visualization of synthetic datasets generated by MPS2U for MNIST, FashionMNIST, CIFAR-10, and CIFAR-100.

Performance Metrics (MPS2U):

The results for the MPS2U model, evaluated on four datasets, are as follows:

- **MNIST:**
MPS2U achieves an impressive accuracy of 95% using only 50 images per class (ipc). This demonstrates that the model is effectively able to distill the key features of the MNIST dataset while significantly reducing its size.
- **FashionMNIST:**
For FashionMNIST, MPS2U reaches 77% accuracy with 50 ipc. While slightly lower than the performance on MNIST, the model still maintains a high level of accuracy, effectively reducing the dataset size without sacrificing important features.
- **CIFAR-10:**
The model achieves 31% accuracy on CIFAR-10 with 50 ipc. This lower accuracy is expected due to the increased complexity of CIFAR-10, which contains more varied and challenging image classes compared to MNIST and FashionMNIST.
- **CIFAR-100:**
On CIFAR-100, MPS2U attains 12% accuracy with 50 ipc. The drop in performance on this dataset reflects its increased complexity, where the model struggles to preserve essential information with a reduced dataset.
- **Average Training Time:**
MPS2U demonstrates efficient training, with an average time of approximately 5 minutes across all four datasets. This highlights the advantage of using synthetic datasets, significantly reducing the training time without a substantial loss in accuracy.

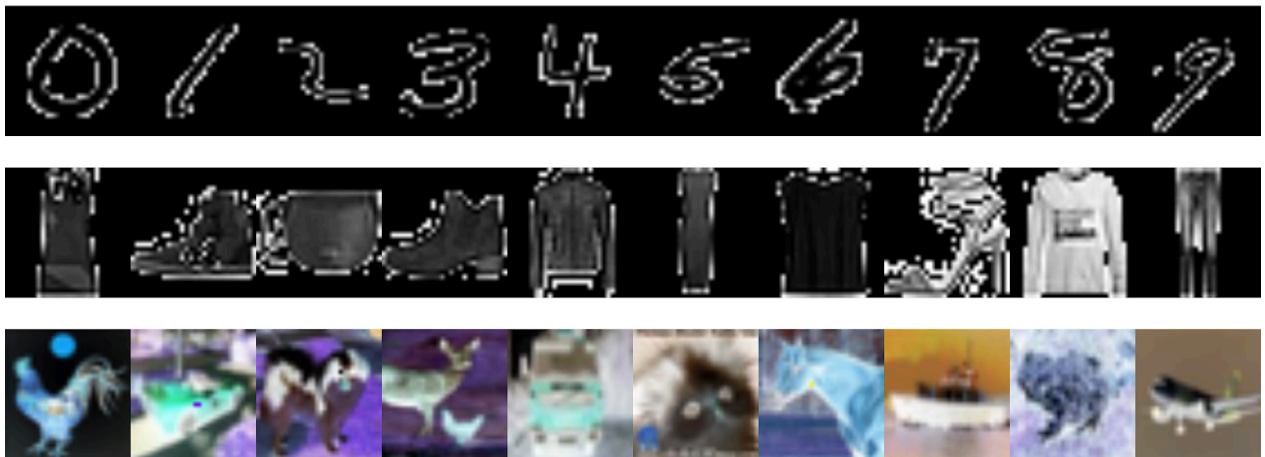


Figure 3: Visualization of synthetic datasets generated by Mutual Info for MNIST, FashionMNIST, and CIFAR-10

Performance Metrics (Mutual Information):

The results for the Mutual Information (Mutual Info) model, tested on three datasets, are as follows:

- **MNIST:**
The Mutual Info model achieves 88% accuracy with 50 ipc. This is slightly lower than MPS2U's performance but still reflects good efficiency in generating synthetic datasets.
- **FashionMNIST:**
For FashionMNIST, Mutual Info maintains 77% accuracy with 50 ipc, which matches MPS2U's performance, indicating that the Mutual Info approach is also effective for moderately complex datasets.
- **CIFAR-10:**
On CIFAR-10, the Mutual Info model records 23% accuracy with 50 ipc. Similar to MPS2U, the model struggles with the complexity of CIFAR-10, showing a notable decline in accuracy compared to simpler datasets.
- **Average Training Time:**
Mutual Info requires more training time, averaging approximately 10 minutes across the three datasets. This increased training time suggests that the Mutual Info approach is more computationally intensive than MPS2U.

This study shows that synthetic dataset generation methods, MPS2U and Mutual Information, can effectively reduce dataset size while maintaining good accuracy, especially on simpler datasets like MNIST and FashionMNIST. MPS2U is efficient with fast training times, whereas Mutual Info, though more accurate on MNIST, requires more computational resources. While both methods work well for simpler datasets, challenges remain with complex datasets like CIFAR-10 and CIFAR-100, where accuracy drops.

6. CONCLUSION

Through models like MPS2U and MPS2U with Mutual Information, we achieved significant accuracy with minimal computational power. Reinforcement learning played a key role in generating synthetic data and significantly boosting model efficiency, reducing the need for large datasets and enabling faster training times without sacrificing performance.

While the results on simpler datasets like MNIST and FashionMNIST are promising, there is still room to improve accuracy on more complex datasets like CIFAR-10 and CIFAR-100. Future work could focus on refining the synthetic dataset generation process to handle these challenges better. Additionally, improving the implementation of Mutual Information could enhance the selection of the most informative data points, ensuring better model performance with reduced data.

In conclusion, this study demonstrates the potential of synthetic datasets for training, with opportunities for further optimization, particularly for more complex tasks.

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APPENDIX

GitHub Link: <https://github.com/likith-sg/Dataset-Condensation>